

ELLSWORTH AFB, SD, USA

WMO: 726625

Lat: 44.150N

Lon: 103.100W

Elev: 3278

StdP: 13.04

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24006

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.5	-2.3	-12.7	3.1	-5.9	-7.9	4.1	-1.3	41.9	29.1	38.6	25.0	8.9	10	0.657

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.4	95.3	66.6	91.3	66.0	88.2	65.6	72.0	84.7	70.2	83.4	68.5	81.9	11.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.5	118.4	76.7	66.1	108.9	75.9	64.0	101.0	74.5	37.9	84.2	36.2	83.2	34.6	82.0	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
35.3	30.2	26.0	-13.6	101.2	5.7	3.8	-17.7	103.9	-21.1	106.1	-24.3	108.2	-28.5	111.0		
			-14.0	75.5	5.6	3.7	-18.0	78.2	-21.3	80.4	-24.4	82.5	-28.5	85.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.8	25.1	27.2	35.8	44.5	54.6	65.9	73.2	72.4	63.3	48.2	36.0	25.6
	DBStd	20.30	13.23	14.03	13.41	10.04	8.52	7.65	6.07	6.17	9.16	9.90	12.13	13.07
	HDD50	3497	772	642	459	217	49	3	0	0	12	149	436	758
	HDD65	7002	1236	1059	904	616	335	78	8	13	135	524	871	1223
	CDD50	2683	1	3	20	53	192	479	719	694	411	94	16	1
	CDD65	712	0	0	0	2	14	104	262	242	84	4	0	0
	CDH74	8425	0	0	16	63	225	1245	2897	2676	1179	116	8	0
	CDH80	3628	0	0	1	11	58	485	1320	1242	479	31	1	0
	WSAvg	10.8	11.2	11.6	11.8	12.5	11.5	10.2	9.1	9.3	9.9	11.3	10.5	10.6
	PrecAvg	17.7	0.4	0.4	0.9	2.2	3.5	2.9	2.2	1.9	1.3	1.5	0.5	0.3
Precipitation	PrecMax	23.7	1.0	0.9	1.9	4.8	7.3	6.3	3.9	4.2	3.0	5.4	1.3	1.1
	PrecMin	12.2	0.0	0.0	0.0	0.4	1.5	0.7	0.6	0.8	0.2	0.1	0.0	0.0
	PrecStd	3.5	0.3	0.3	0.5	1.0	1.7	1.6	0.9	0.9	0.7	1.3	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.3	67.5	75.6	81.4	86.3	98.3	99.5	99.6	97.4	84.5	73.2	61.8
		MCWB	43.0	47.3	51.0	55.2	63.0	65.6	67.7	66.7	64.7	57.8	50.0	44.3
	2%	DB	53.4	58.7	68.1	73.7	80.1	90.1	94.9	94.9	90.1	76.9	64.8	54.0
		MCWB	41.2	43.1	48.6	53.2	60.2	65.9	68.2	65.9	63.0	54.7	46.5	40.9
	5%	DB	48.2	52.3	61.8	67.7	75.0	85.8	91.1	90.9	85.7	70.4	59.2	47.6
		MCWB	38.2	40.2	45.6	51.1	58.1	65.5	67.4	65.5	62.0	52.2	44.2	37.5
	10%	DB	43.7	46.0	55.7	62.2	70.7	81.3	87.6	87.4	81.0	64.5	53.9	42.7
		MCWB	35.6	36.8	43.0	48.3	56.6	64.1	67.1	65.6	59.9	50.1	41.9	34.7
	0.4%	WB	44.7	47.9	53.1	57.7	67.1	73.5	75.4	73.9	69.6	60.1	51.0	45.7
		MCDB	57.8	65.9	71.6	74.5	79.4	84.3	85.4	84.9	84.1	77.5	72.0	61.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.2	44.0	49.2	54.3	63.0	70.0	72.6	70.9	66.3	56.7	47.5	41.3
		MCDB	51.9	57.6	66.9	70.6	75.5	82.3	85.8	83.8	81.8	71.8	63.2	52.7
	5%	WB	38.7	40.9	46.3	51.8	60.1	67.8	71.0	68.9	64.1	53.5	45.0	38.0
		MCDB	47.2	51.0	60.6	65.4	72.0	81.1	84.2	83.1	80.5	67.3	58.2	46.6
	10%	WB	36.1	37.5	43.5	49.2	57.7	65.7	69.3	67.3	61.8	51.0	42.1	35.1
		MCDB	43.5	45.4	54.8	60.8	67.5	78.9	82.3	81.6	77.5	63.5	53.1	42.0
Mean Daily Temperature Range	5% DB	MDBR	20.7	21.0	22.6	22.2	21.4	23.2	24.4	25.7	26.0	22.7	21.9	19.8
		MCDBR	27.3	29.8	33.2	33.1	29.2	30.4	29.7	31.5	33.4	32.5	30.7	26.9
		MCWBR	17.4	17.4	17.6	17.0	14.9	13.4	11.1	11.8	13.0	15.6	16.8	16.6
	5% WB	MCDBR	26.1	28.0	31.0	29.8	26.1	27.0	24.9	26.7	28.8	28.8	29.3	25.6
		MCWBR	17.2	16.7	17.0	16.1	14.0	14.1	11.9	12.0	12.1	14.6	16.6	16.5
Clear-Sky Solar Irradiance	taub		0.252	0.261	0.279	0.313	0.331	0.344	0.366	0.361	0.325	0.288	0.262	0.251
	taud		2.579	2.552	2.552	2.441	2.467	2.456	2.425	2.419	2.504	2.578	2.565	2.564
	Ebn at Noon		288	302	308	302	298	293	285	283	287	287	280	277
	Edh at Noon		20	24	28	34	34	35	36	34	29	23	20	18
All-Sky Solar Radiation	RadAvg		607	914	1283	1591	1889	2193	2233	1920	1517	996	671	507
	RadStd		31	53	100	94	184	163	82	85	94	95	51	37

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page